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Interaction of Asthenospheric and Lithospheric Mantle: The Genesis of Calc-alkaline Volcanism at Erciyes Volcano, Central Anatolia, Turkey

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Abstract

The quaternary Erciyes stratovolcano and its associated monogenetic cones exhibit tholeiitic, alkaline, and calc-alkaline suites, with the calc-alkaline series dominating. Calc-alkaline products are andesitic, dacitic, and rhyodacitic, showing moderate to low MgO and high K₂O, plus LIL and HFS element contents. The andesites display trace-element patterns similar to those of the Central Mexican Volcanic belt, the Colville igneous complex (northeastern Washington, USA) and the Taos Plateau volcanic field (in north-central New Mexico, associated with northern Rio Grande Rift). Dacites from Erciyes also resemble these igneous complexes and give patterns similar to those of the upper crust. Furthermore, the high Ba/Nb ratio (> 28) is typical of arc volcanism (Fitton et al., 1988) ranging from 27 to 47 for andesites, and 27 to 50 for dacites at Erciyes. Calc-alkaline suites also show high Rb/Nb ratios (4–7.5). Moreover, ⁸⁷Sr/⁸⁶Sr and ¹⁴³Nd/¹⁴⁴Nd ratios range from 0.703434 to 0.705143, and 0.51294 to 0.5126 for andesites and dacites, respectively. These incompatible element and isotopic ratios suggest two possible mechanisms for the genesis of calc-alkaline suites: subduction enrichment or crustal involvement.

Isotopic similarities between tholeiitic and andesitic rocks, LIL and HFS element compositions, and spider plots all indicate that various source components have been involved in the genesis of these products. The calc-alkaline products of Erciyes volcano were generated by the mixing of partial melts derived from depleted (tholeiitic) and enriched asthenospheric sources. In addition, the calc-alkaline suite was derived from an initially tholeiitic basalt end-member via a 25–30% assimilation of crustal material. Thus, post-collisional extension led to development of the Erciyes volcanism in central Anatolia. Calc-alkaline products were widely generated by the combined effects of mixing of depleted and enriched asthenospheric sources, together with a progressive AFC process that started from an initially tholeiitic component, without direct melting of the pre-subducting slab.

Introduction

THE DEVELOPMENT OF volcanic systems within the Anatolian block can be linked to continental collision between the Afro-Arabian and Eurasian plates during the Early and Middle Miocene (Şengör, 1980; Şengör and Yilmaz, 1981; Dewey et al., 1986). This convergence resulted in the Anatolian

block moving westward along the Northern (NAF) and Eastern Anatolian (EAF) strike-slip faults (McKenzie, 1972; Şengör, 1980; Jackson and McKenzie, 1984; Dewey et al., 1986; Chorowicz et al., 1994). Westerly motion of the Anatolian block was also confirmed by Oral et al. (1994), Le Pichon et al. (1995), and Yürür and Chorowicz, (1998) by means of space-geodetic data analysis (Fig. 1). This movement has also been linked to internal deformation of the central part of Anatolian block, as represented by pull-apart basins and thrust faults (Şengör

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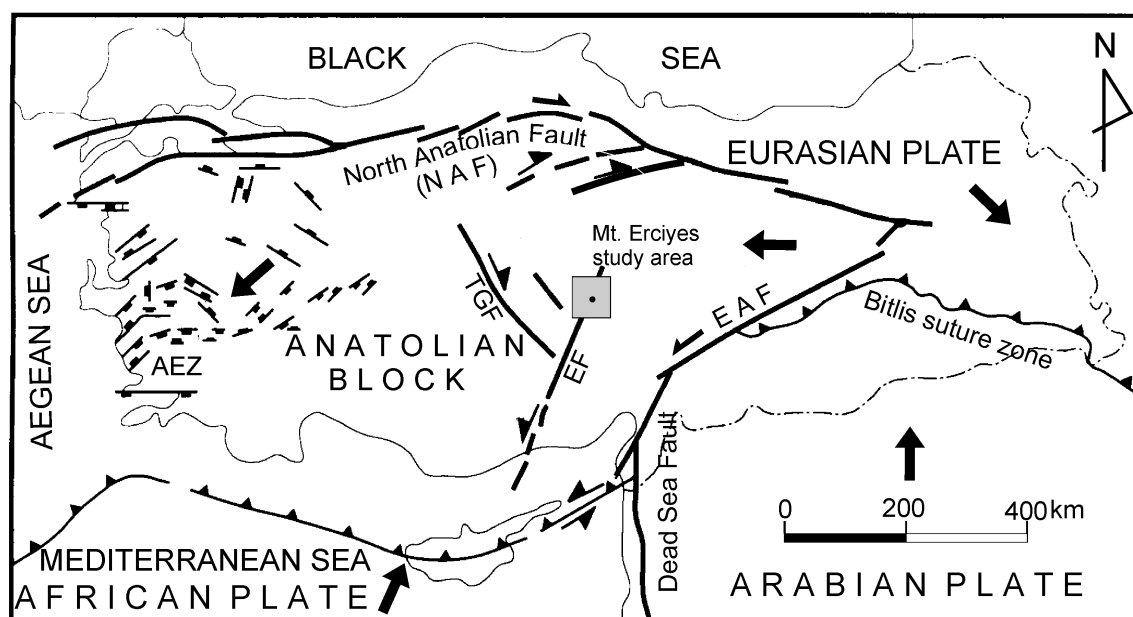


FIG. 1. Neotectonic map and major tectonic structures of the Anatolian block, modified from Bozkurt and Koçyigit (1983). Abbreviations: AEZ = Aegean Extension Zone; NAF = North Anatolian Fault; EAF = East Anatolian Fault; EF = Ecemiş Fault; TGF = Tuz Gölü Fault. Black arrows indicate the direction of plate motion; black half-arrows show the relative direction of motion on the faults.

et al., 1985; Kempler and Garfunkel, 1991; Lyberis et al., 1992).

As a result of such neotectonic change, the Anatolian block was subject to the combined effects of both subduction and extension. Moreover, subduction-related tectonics occurred largely in the Aegean region, whereas collision tectonics was generally responsible for the volcanic evolution of other sections of plate boundary (Fig 1). In both regions, highly contaminated calc-alkaline magmas were followed by uncontaminated alkaline magmas during the final stage of the compressional regime, during the Middle Miocene in the west and the Quaternary in the east (e.g., Yilmaz, 1990; Deniel et al., 1998).

Magmatic development is a complex interaction of processes governed by the composition of source components and host rocks, in addition to the composition and physical and thermal properties of the extracted magma. Variable volumes of differentiated lava products (basaltic andesite, andesite, dacite, and rhyodacite) erupt periodically throughout the long lifetime of large calc-alkaline magmatic systems.

Alkaline and calc-alkaline volcanism developed in the central part of the Anatolian block. Quaternary Erciyes volcanism exhibits tholeiitic, alkaline, and calc-alkaline basaltic associations, with calc-

alkaline lavas dominating, such as basalt, andesite, dacite, and rhyodacite.

Erciyes basaltic products are tholeiitic and alkaline in composition, and were generated from depleted and partly enriched asthenospheric mantle components (Kürkçüoğlu, 1998, 2001). Alternative hypotheses relate to subduction (Pasquare, 1968; Innocenti et al., 1975; Ayranci, 1991) or intraplate processes (Aydar, et al., 1995; Notsu et al., 1995, Kürkçüoğlu, 1998, 2001).

In this study, we present new trace- and rare-earth-element (REE) data together with Sr and Nd isotopic ratios and modeling of the probable geochemical processes to provide further information on the possible interaction of lithospheric and asthenospheric sources beneath central Anatolia. Other options for the genesis of evolved calc-alkaline products found at Erciyes include a direct (partial melting) or indirect (fluid transport to the mantle wedge) contribution from the subducting plate, the partial melting of crustal and asthenospheric sources, and a lithospheric contribution.

Geological Background

Central Anatolia exhibits various types of volcanics such as ignimbrites, monogenetic vent products

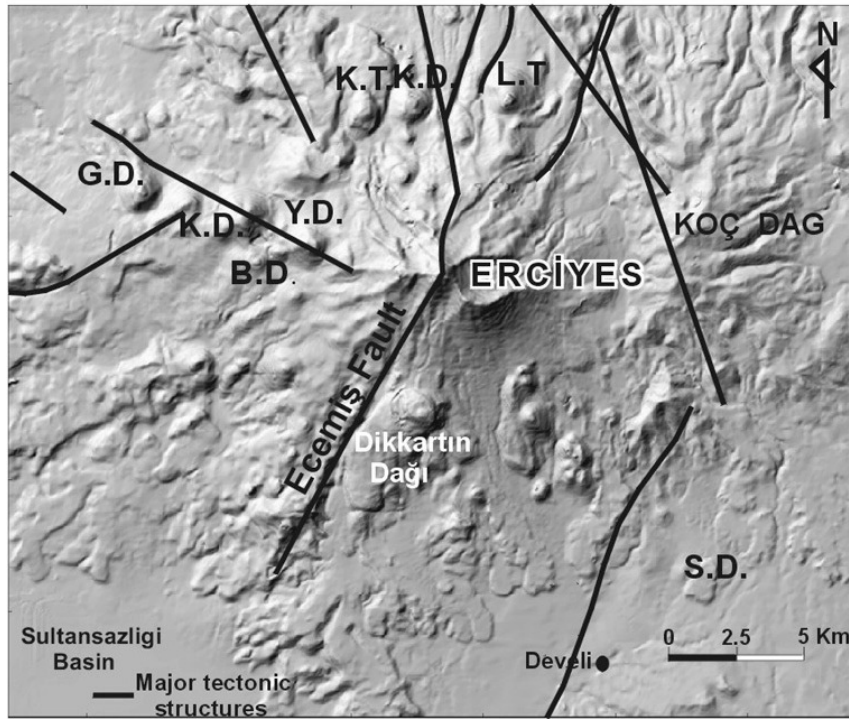


FIG. 2. Digital Elevation Model (DEM) of Mt. Erciyes stratovolcano and locations of major tectonic structures. The DEM was made by digitizing the elevation contours of topographic maps of a scale of 1:25,000 and artificially illuminated from the north. Vertical exaggeration is $\times 1.5$. Abbreviations (from left to right): G.D. = Gög Dag; K.D. = Kolanlı Dag; B.D. = Boz Dag; Y.D. = Yilband Dag; K.T. = Karagüllü Tepe; K.D. = Kefeli Dag; L.T. = Lifos Tepe; S.D. = Sultan Dag.

(scoria cones and associated lava flows, maars, and domes), and some stratovolcanoes (Melendiz, Hasan, Erciyes). At 3917 m, Mt. Erciyes, located almost 15 km south of the city of Kayseri, is the highest stratovolcano of Central Anatolia (Fig. 2). The volcano mainly exhibits lava flows that relate to central and adventive monogenetic vents (domes and scoria cones) and their associated pyroclastic products. Two main evolutionary stages can be seen (Şen, 1997; Şen et al., 1999). Volcanic activity began with basaltic lava flows, tholeiitic in composition, followed by differentiated sequences (basaltic andesite, andesite, dacite, and rhyodacite), and terminating with extensive ignimbritic eruptions (Şen et al., 1999, 2002), dated by Innocenti et al. (1975) using the K/Ar method (2.7 ± 0.1 , 2.8 ± 0.1 , and 3.0 ± 0.1 Ma). The second stage involved basaltic andesitic, andesitic, dacitic, and rhyolitic lavas, and also terminated with pyroclastic eruptions and debris avalanches. Previous studies indicate that Erciyes volcanism is a consequence of post-collisional extension in an intraplate tectonic environment (Aydar, et al., 1995; Notsu et al., 1995; Kürkcüoğlu et al., 1998, 2001).

Detailed geochemical studies have been carried out by Kürkcüoğlu et al. (1998, 2001). The authors suggest that the tholeiitic basalts were derived from a primitive mantle source via 1–1.5% fractional melting, with no crustal involvement. Calc-alkaline basalts were also produced by 10–15% crustal assimilation from initially tholeiitic sources. The combined effects of partial melting and AFC processes are thought to be responsible for the genesis of the calc-alkaline suite (Kürkcüoğlu, 2000).

Geochemistry

Analytical techniques

Major and trace element analyses were performed at the Geological Engineering Department of Hacettepe University, Turkey, using a Philips PW 1480 X-Ray Fluorescence spectrometer, from fused glass tablets and pressed powdered samples. A detailed explanation of sample preparation was given by Alici et al. (1998). The spectrometer was calibrated using international standards (USGS and GEOSTANDARTS) to ensure accurate data. Rare earth element analyses were performed using a

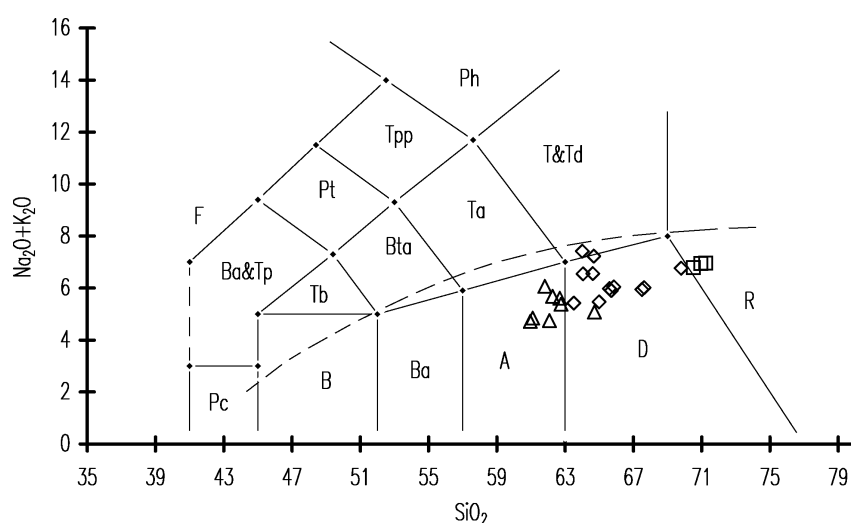


FIG. 3. Total alkaline versus silica diagram (Le Bas et al., 1986) of Erciyes calc-alkaline (andesite, dacite, and rhyodacite) products. The dashed line separates alkaline and sub-alkaline fields, after Miyashiro (1978).

JY7011 Inductively Coupled Plasma Emission (ICP) spectrometer at Blaise Pascal University in Clermont-Ferrand, France. Sr and Nd isotopic compositions were determined by means of a fully automated V.G. isomass 54E and Thompson-CSF TSN 206SA mass spectrometers at Blaise Pascal University in Clermont-Ferrand, and at the Geological Engineering Department of Hacettepe University, Ankara (Turkey). U, Pb, Th, Hf, and Ta concentrations were determined using an Inductively Coupled Plasma-Mass Spectrometer (ICP-MS) at ACME Analytical Laboratories Ltd. in Canada. Analytical errors are typically $\pm 2\%$ for major elements and less than $\pm 10\%$ for trace elements. $^{87}\text{Sr}/^{86}\text{Sr}$ ratios were normalized to $^{86}\text{Sr}/^{88}\text{Sr} = 0.1194$. $^{143}\text{Nd}/^{144}\text{Nd}$ ratios were normalized to $^{146}\text{Nd}/^{144}\text{Nd} = 0.7219$. Over the measurement periods, replicate standards analyses gave, for the National Bureau of Standards (NBS) 987, $^{87}\text{Sr}/^{86}\text{Sr} = 0.710244 \pm 4$ (2σ), and for La Jolla $^{143}\text{Nd}/^{144}\text{Nd} = 0.51184 \pm 3$ (2σ).

Major-element geochemistry

The SiO_2 content of the calc-alkaline lavas ranges from 60 to 71 wt%. The major-element compositions of the calc-alkaline suite are listed in Table 1 and plotted on a total alkaline versus silica (Le Bas et al., 1986) diagram (Fig. 3). The lavas exhibit andesitic, dacitic, trachydacitic, and rhyolitic compositions. Oxide versus SiO_2 diagrams show that Al_2O_3 , Fe_2O_3 , MnO , MgO , CaO , TiO_2 , and P_2O_5 correlate linearly and negatively, whereas K_2O

and Na_2O correlate linearly and positively with increasing SiO_2 . Such trends are related to fractional crystallization with separation of plagioclase and ferromagnesian minerals.

Trace-element and REE geochemistry

Trace-element and REE compositions are listed in Tables 1 and 2. SiO_2 versus incompatible-element diagrams show that Ba and Rb increase whereas Sr decreases with increasing SiO_2 . Although these elements reflect classical evolution trends, the behavior of other incompatible elements is unusual. Despite being highly incompatible elements, Zr, Nb, and Y do not show positive correlations with increasing silica. Fractional crystallization cannot provide a unique explanation for such variations.

In multi-element diagrams, MORB-normalized (Pearce, 1982) patterns of the calc-alkaline suite display a signature that closely reflects subduction, but differs in terms of high-field strength (HFS) element enrichment and minor depletion of Nb (Figs. 4A–4D). Furthermore, the andesites exhibit trace-element patterns that resemble those of the Central Mexican Volcanic Belt, the Colville igneous complex (northeastern Washington, USA) and the Taos Plateau volcanic field (USA) (Figs. 4B–4C). Erciyes dacites also resemble the Taos volcanics and the Colville igneous complex, and show patterns similar to those of upper crustal values (Figs. 5A–5B). Verma (1999) suggested that andesitic magmas from

TABLE 1. Major (%) and Trace-Element (ppm) Compositions, and Sr, Nd Isotopic Ratios of Representative Calc-alkaline Products¹

	Andesite								Dacite								Rhyodacite			
	ERC96-109	ERC95-128	ERC98-3	C88-80	ERC96-151	ERC95-8	ERC95-9	ERC96-71	ERC91-21	ERC91-27	ERC95-62	ERC95-111	ERC96-55	ERC95-14	ERC95-75	ERC91-1	ERC92-1	ERC92-3	ERC91-28	
SiO ₂	61.45	61.02	62.52	61.89	60.45	60.24	63.34	64.72	65.51	68.87	66.88	65.84	63.51	62.25	63.72	63.24	65.06	70.1	70.54	
Al ₂ O ₃	17.25	15.95	16.58	16.54	17.58	17.45	16.38	16.56	16.21	14.77	15.91	15.5	16.35	15.75	16.07	16.47	16.12	14.78	15.21	
Fe ₂ O ₃	5.05	5.63	5.08	5	5.39	4.93	4.76	4.88	4.06	3.19	3.82	3.62	4.38	5.17	5.27	4.89	4.31	2.68	2.81	
MnO	0.078	0.093	0.082	0.08	0.089	0.079	0.065	0.081	0.07	0.06	0.061	0.062	0.074	0.082	0.081	0.08	0.07	0.05	0.06	
MgO	3.04	3.03	3.37	3.19	3.26	2.22	2.19	2.02	1.97	1.28	1.69	1.6	2.04	1.55	1.23	2.78	2.21	0.92	0.96	
CaO	6.58	5.7	5.52	6	6.84	5.65	4.84	4.31	4.91	3.32	4.22	4.29	5.32	4.08	3.85	5.91	4.71	3	3.21	
Na ₂ O	3.34	3.72	3.83	3.52	3.44	3.75	4.1	4.54	3.93	3.85	3.72	3.6	3.55	4.2	4.6	3.8	3.78	3.9	4	
K ₂ O	1.36	1.85	1.76	1.78	1.23	2.16	2.37	2.03	2.08	2.82	2.17	2.26	1.79	3.01	2.52	1.6	2.15	2.96	2.79	
TiO ₂	0.67	0.8	0.74	0.62	0.68	0.79	0.69	0.78	0.58	0.4	0.5	0.47	0.58	0.93	0.95	0.63	0.6	0.33	0.37	
P ₂ O ₅	0.18	0.22	0.25	0	0.2	0.2	0.16	0.24	0.18	0.13	0.14	0.13	0.15	0.25	0.24	0.19	0.19	0.1	0.11	
LOI	0.54	0	0.09	1.03	0.46	0.75	0.44	0.31	0.09	1.04	0.54	0.02	0.84	0.99	1.28	0.8	0.99	2.01	0.29	
Total	99.54	98.01	99.82	99.65	99.63	98.21	99.33	100.48	99.59	99.73	99.64	97.38	98.57	98.26	99.82	100.39	100.19	100.83	100.35	
Nb	10.2	14.2	7.08	5.1	13.4	10.22	15.8	10.31	14.5	17	13.3	12.1	10	17.7	17.8	15.8	15.8	15.6	15.7	
Zr	171	204.5	153.6	127.3	149.3	157.8	246.9	228.8	185.7	153.5	177.4	167.8	167.5	331.2	332.4	184.9	175.1	148.9	171.7	
Y	21	26.9	16.2	23.3	18.7	16.7	27.7	20.4	27.3	25.6	24.3	24.3	22.1	36.4	38.7	25.2	24.3	23.8	25	
Sr	335.4	323.3	291	401.9	329.7	292.1	285.8	264.5	295.2	210.3	250.4	248.8	280.9	298.1	296.8	314.9	275.4	194.5	210.1	
Rb	37.1	61.3	59.32	1.7	35.7	58.17	79.7	69.42	84.9	113.7	73.7	90.6	65	88	87	55.2	92.2	146.5	133.7	
Ga	18.4	19	15.1	17	18.9	16.6	19.1	15.7	18.3	17.5	17.4	17.8	18.2	21.5	22.1	18.7	18.4	16.5	16.4	
Ni	38.2	32.5	7	111.1	26	8	18.4	12	23.1	12	17.4	12.1	15	8.1	10.5	31	23.5	7.2	8.2	
Co	44	25.8	29.7	44.7	39	20.8	24.8	26.3	13.2	7.5	20.2	17.1	27.8	22.6	32	16.2	14.8	6.2	5.8	
Cr	65.9	50.9		215.6	38.7		14.3		68.4	72.8	23.8	12.8	17.6	6.9	5.2	119.9	61.2	52.1	48.1	
V	90.9	94.8	78	164.3	88.9	86	70.9	72	69.4	37.9	48.5	49.4	69	78.4	85.6	81.4	55.2	28.6	28.9	
Ba	289.7	387.9	337.6	59.2	257.8	404.5	481.8	427	396.8	492.2	408.7	429.4	371.7	486.8	522.3	339.6	469.3	510.5	482.8	
⁸⁶ Sr/	0.704866			0.703434		0.704868		0.70461 ± 0.705143										0.705474 ± 10	0.705414 ± 12	
⁸⁷ Sr	± 10			± 13		± 10		9	± 12									± 10	± 12	
¹⁴³ Nd/	0.512652			0.512942		0.512631		0.512611 ± 13	0.5126									0.512598 ± 8	0.512542 ± 10	
¹⁴⁴ Nd	± 7			± 13		± 9		± 13	± 7									± 8	± 10	

¹Andesite, dacite, rhyodacite.

TABLE 2. Rare-Earth and Th, U, Pb, Hf, and Ta Compositions of Selected Samples, ppm.

	ERC98-3	ERC95-8	ERC96-71	ERC91-21	ERC92-3
La	19.5	24.5	26.2	25.9	28.5
Ce	36.9	47.8	47.2	48.5	53.9
Pr	4.1	n.a	5.61	n.a	n.a
Nd	16.7	17.9	22.8	19.1	17.5
Sm	3.9	3.83	4.6	3.81	3.33
Eu	0.95	0.93	1.09	0.83	0.61
Gd	4.24	3.61	5.34	3.77	3.1
Dy	3.3	3.42	4.23	3.55	3.05
Er	1.96	1.86	2.42	1.96	1.74
Yb	1.83	1.98	2.21	2.1	2.1
Lu	0.23	0.3	0.27	0.33	0.33
Th	10.7	10.3	12.2		
Pb	6	5	6		
U	2.7	2.8	3.2		
Hf	4.2	4.5	6		
Ta	1	1.3	1.3		

the Central Mexican Volcanic Belt have a crustal origin, and were generated by partial melting of a heterogeneous mafic granulite source in the lower crust. The Colville igneous complex formed as a result of decompression melting of crust and mantle (Morris et al., 2000). In addition, chondrite-normalized (Nakamura, 1974) REE patterns show LREE enrichment in all phases and a small negative Eu anomaly in dacitic compositions. REE spidergrams also display patterns similar to those of the Central Mexican Volcanic Belt and the Colville igneous complex (Figs. 5C–5D). The similarity of multi-element patterns between products from Erciyes and other igneous provinces suggests the possibility of similar sources during the generation of the calc-alkaline products.

Isotope geochemistry

$^{87}\text{Sr}/^{86}\text{Sr}$ and $^{143}\text{Nd}/^{144}\text{Nd}$ isotopic ratios range from 0.703434 to 0.704998 and 0.512942 to 0.512631; 0.705143 to 0.70461 and 0.512611 to 0.512594; and 0.705474 to 0.705414 and 0.512598

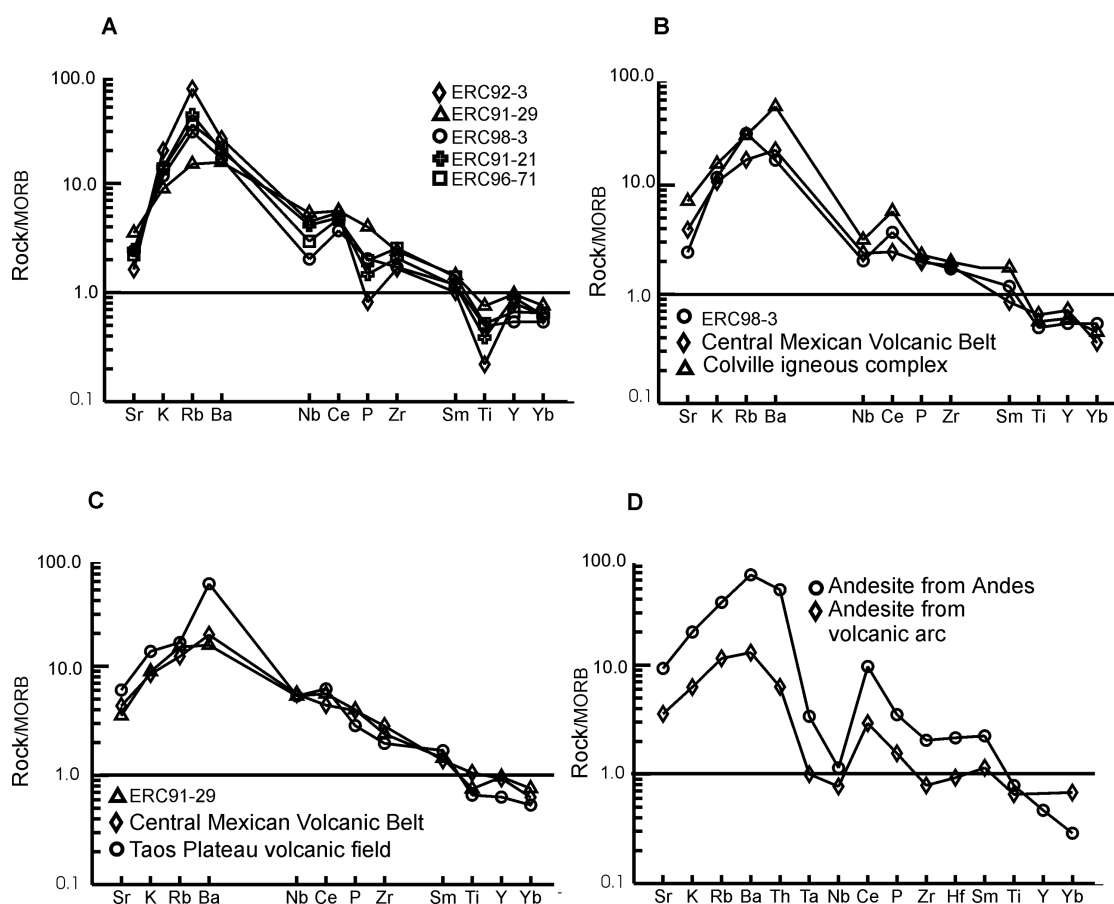


FIG. 4. MORB-normalized (Pearce, 1983) trace-element patterns for representative andesitic samples of Erciyes volcanics. References are from Pearce (1982), McMillan and Dungan (1986), Davidson et al. (1990), Verma (1999), and Morris et al. (2000).

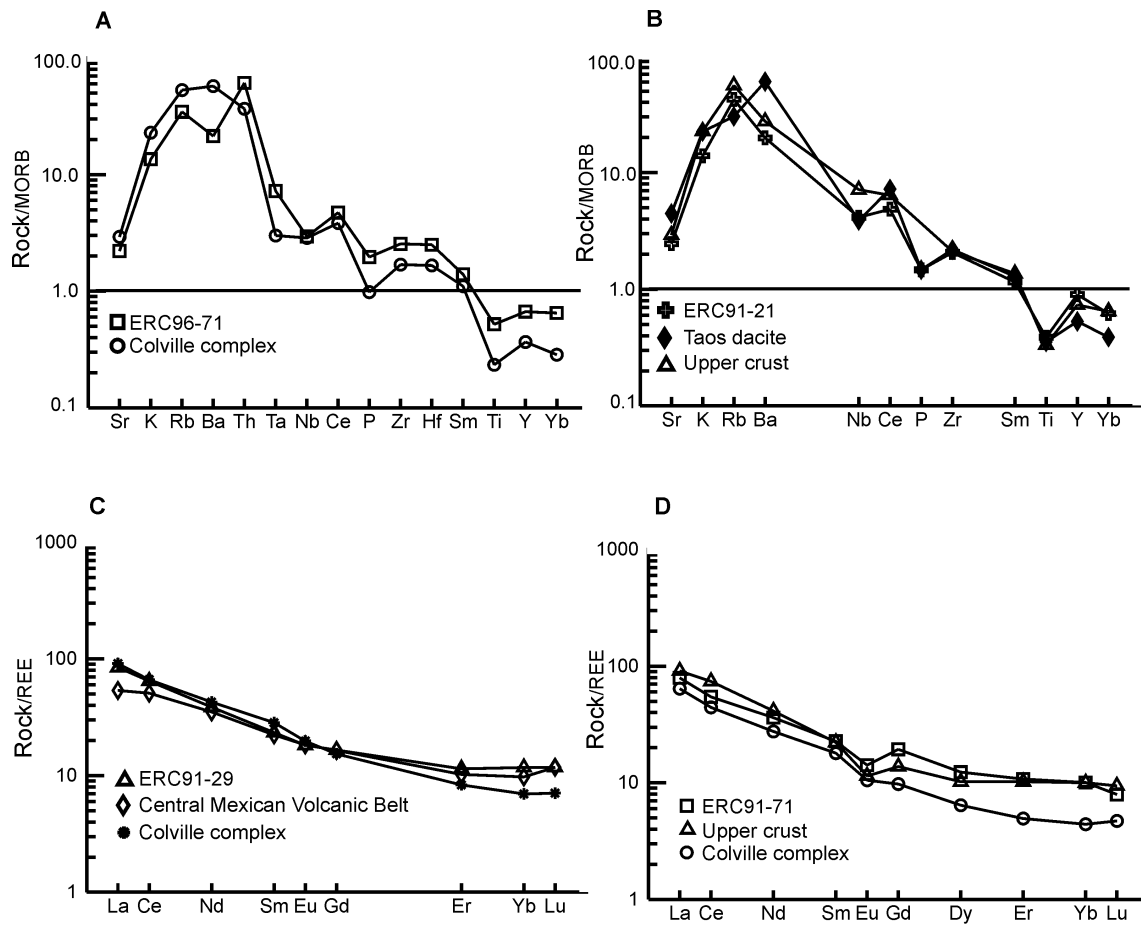


FIG. 5. MORB-normalized (Pearce, 1983) trace-element patterns (A and B) and chondrite-normalized REE patterns (C and D) for representative dacitic samples of Erciyes volcanics. References are from Taylor and McLennan (1985), McMillan and Dungan (1986), Verma (1999), and Morris et al. (2000).

to 0.512542 for andesites, dacites, and rhyodacites respectively (Table 1). The $^{87}\text{Sr}/^{86}\text{Sr} - ^{143}\text{Nd}/^{144}\text{Nd}$ diagram (Fig. 6) shows that andesitic products plot within the mantle array, close to the Rio Grande Rift axis and the Central Mexican Volcanic Belt, and one sample plots in the Basin and Range province. Moreover, the dacitic rocks plot at the lower end of the mantle array field, whereas the more siliceous rhyodacites fall in the enriched field of the diagram. The $^{87}\text{Sr}/^{86}\text{Sr} - ^{143}\text{Nd}/^{144}\text{Nd}$ diagram (Fig. 6) suggests asthenospheric sources and involvement crustal materials for the origin of calc-alkaline lava suites. The andesites that fall in the Basin and Range province field have $^{87}\text{Sr}/^{86}\text{Sr}$ and $^{143}\text{Nd}/^{144}\text{Nd}$ isotopic ratios and incompatible-element contents that compare with those of the Erciyes' tholeiitic basalts (Kürkçüoğlu, 2000; Kürkçüoğlu et al., 2001). Thus, both approaches suggest that the tholeiites may be one of the agents responsible for

the development of the calc-alkaline suite. Also, the rhyodacites fall in the enriched field of the $^{87}\text{Sr}/^{86}\text{Sr} - ^{143}\text{Nd}/^{144}\text{Nd}$ diagram, and may be considered as the more siliceous differentiates of the mafic tholeiitic end-member. This hypothesis is also supported by the large-ion lithophile (LIL) element concentrations of the siliceous samples.

Discussion and Conclusions

Recent studies demonstrate that calc-alkaline lavas were generated from various source components that were affected by differing geochemical processes. The conventional process responsible for producing calc-alkaline lavas is differentiation of a basaltic source magma with or without crustal involvement. Direct partial melting from the mantle wedge also produces calc-alkaline magma (Ringwood, 1990); it may be generated directly by partial

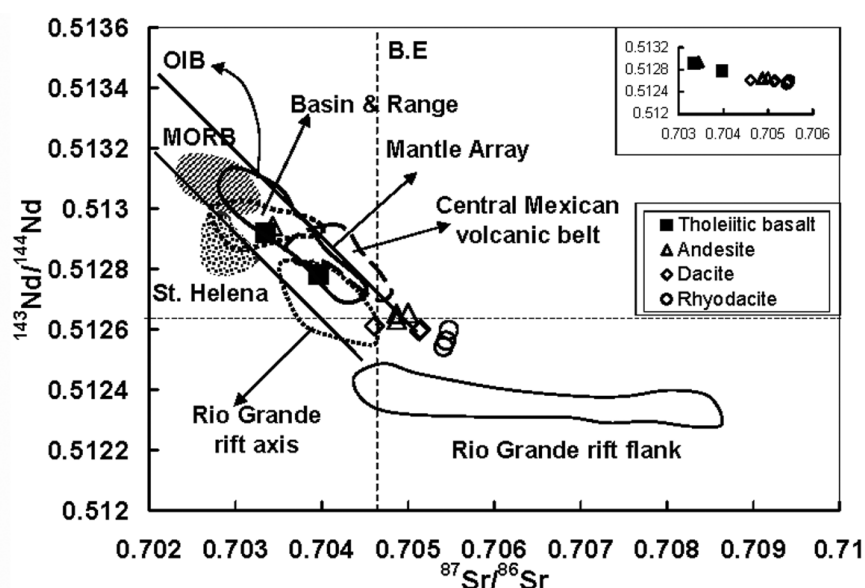


FIG. 6. $^{87}\text{Sr}/^{86}\text{Sr}$ – $^{143}\text{Nd}/^{144}\text{Nd}$ diagram for selected calc-alkaline samples of Erciyes volcanics. Data are from Cohen and O'Nions (1982), Chen and Frey (1985), Chaffey et al. (1989), Chen et al. (1990), Kempton et al. (1991), Gibson et al. (1993), and Verma (1999). Mantle array and bulk-earth (B.E) are from Faure (1986).

melting of hydrous lherzolite at depths of less than 40 km (Wilson, 1989). Takagi et al. (1999) argued that rhyolitic magma may be generated from a metasomatized mantle wedge via a low degree of partial melting in subduction zones. Verma (1999) proposed that calc-alkaline magma is generated by partial melting of the lower crust. The following sections consider the origin of the Erciyes calc-alkaline suite.

Parental compositions of andesite, dacite, and rhyodacites

Taken together, isotopic similarities between tholeiitic and andesitic rocks, LIL and HFS element compositions, and spider element plots, all suggest that there are various possibilities for the genesis of the Erciyes calc-alkaline products: (1) andesites and dacites may be the products of different mantle and crustal source components; (2) source components may have undergone several geochemical processes over the course of the calc-alkaline development; and (3) tholeiites may be a source component for the calc-alkaline suite.

Calc-alkaline volcanism is a common feature of subduction zones (Ringwood, 1990; Wilson, 1989). Its genesis is related to the open-system differentiation of a basaltic parental magma (Leeman, 1986; McMillan and Dungan, 1988). If the Erciyes calc-alkaline suite was produced directly via the partial

melting of a subducted slab, andesitic and dacitic products should have reflected subduction signatures in the spidergrams (Figs. 4A and 4D). But at Erciyes, calc-alkaline lavas differ from subduction signatures in exhibiting weak Nb anomalies, P and Ti depletions, and HFS element enrichment (Fig. 4A). Moreover, $^{87}\text{Sr}/^{86}\text{Sr}$ isotopic ratios for the calc-alkaline suite (Table 1) are not as high as in subduction zones. Furthermore, in central Anatolia, plate tectonic reconstructions for the last 13 m.y. (Lyberis et al., 1992) indicate that the subducted African slab did not reach central Anatolia and thus cannot directly account for the present-day calc-alkaline volcanism in this area (Aydar et al., 1995). These observations rule out the direct effect of subduction. The enrichment of asthenospheric sources by slab-released fluids and melts throughout the development of the calc-alkaline lavas must be taken into account.

Crustal composition is an important factor in the development of the series. Continental crust is represented by high Rb, Th, Ba, K, Sr, and low Nb, Ta, and Zr contents (Taylor and McLennan, 1985). A high Ba/Nb ratio (> 28) is characteristic of arc volcanism, as suggested by Fitton et al. (1988). Ba/Nb ratios for the Erciyes calc-alkaline lavas range from 27 to 47 and 27 to 50 for andesites and dacites, respectively. Also, the high Rb/Nb ratio is typical of continental crust and crustal melts (Pearce et al.,

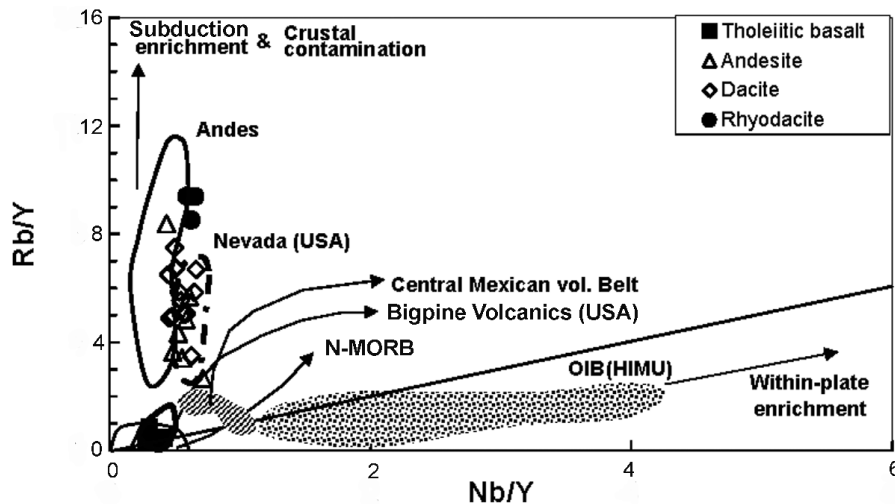


FIG. 7. Nb/Y–Rb/Y diagram of calc-alkaline products. Data are from Le Roex et al. (1983), Chaffey et al. (1989), Davidson et al. (1990), Feeley and Grunder (1991), Ormerod et al. (1991), Woodhead (1996), and Verma (1999).

1984; Weaver and Tarney, 1984) and is about 4.48 (Taylor and McLennan, 1985). Erciyes calc-alkaline lavas also exhibit high Rb/Nb ratios (4–7.5). Additionally, a dacite REE multi-element diagram show pattern that closely resembles crustal signature (Fig. 5D). All the data on incompatible-element ratios and REE plots suggest either partial melting of crustal sources or crustal involvement during the generation of calc-alkaline lavas. The Rb/Y–Nb/Y diagram (Leeman and Hawkesworth, 1986) (Fig. 7) also suggests that the calc-alkaline suite is related to subduction processes or crustal contamination rather than within-plate enrichment.

The $^{87}\text{Sr}/^{86}\text{Sr}$ – $^{143}\text{Nd}/^{144}\text{Nd}$ (Fig. 6) diagram reflects the possible involvement of crustal material. Similar Rb/Nb, Nb/Y, $^{87}\text{Sr}/^{86}\text{Sr}$, and $^{143}\text{Nd}/^{144}\text{Nd}$ (Table 1 and Fig. 7) ratios of the calc-alkaline suite found at Erciyes and the Central Mexican Volcanic Belt imply the involvement of crustal materials in Central Anatolia, able to produce evolved volcanic rocks. High Ba/Nb ratios for the Mexican samples relate directly to the partial melting of heterogeneous lower crust (Verma, 1999) but Ba/Nb ratios for the Erciyes volcanics are not as high. Hence, all geochemical data, incompatible-element contents, LIL/HFS and Sr, Nd isotopic ratios are attributed to a crustal contribution rather than to deep crustal melting.

Assimilation of crustal material and AFC modeling

Small amounts of tholeiite were generated during the development of Erciyes, and exhibit depleted

LIL and HFS element compositions together with low $^{87}\text{Sr}/^{86}\text{Sr}$ and high $^{143}\text{Nd}/^{144}\text{Nd}$ isotopic ratios. Furthermore, the tholeiitic basalts were derived from a primitive mantle source via low degree (1–1.5%) fractional melting in the absence of assimilation or deep crustal melting (Kürkçüoğlu, 2000; Kürkçüoğlu et al., 2001). The fractional melting model was used to calculate the degree of melting.

Erciyes calc-alkaline products may have been generated from an initially tholeiitic end-member via assimilation of lithospheric material and progressive fractional crystallization. The lithospheric contribution is expressed by high Sr and low Nd isotopic compositions: high Ba/Nb, Rb/Nb, Rb/Y, Th/Y, and low Rb/Ba, Nb/Y, and Ba/Rb incompatible-element ratios. An AFC model can be applied starting from an initially tholeiitic source in order to investigate the possible effects of lithospheric involvement.

The chemical variations in primary magma were determined by the assimilation of crustal materials. Chemical changes can be modeled using AFC equations (De Paolo, 1981) based on trace element and Sr, Nd isotopic ratios. For calculation purposes, the tholeiitic sample (C88-79) was selected as a basic end-member due to its depleted composition. The upper continental crust of Taylor and McLennan (1985) was selected as the contaminant. The chemical composition of the contaminant was assumed as Sr = 350 ppm, Nd = 26 ppm (Taylor and McLennan, 1985), and $^{87}\text{Sr}/^{86}\text{Sr} = 0.71600$ (Veizer and Compston, 1974), $^{143}\text{Nd}/^{144}\text{Nd} = 0.51200$ (Rollinson, 1993). Chemical and isotopic compositions,

TABLE 3. $^{87}\text{Sr}/^{86}\text{Sr}$, $^{143}\text{Nd}/^{144}\text{Nd}$ Isotopic Ratios
and Sr and Nd Concentrations Calculated Using AFC Modeling¹

r = 0.1									
F	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
Sr	338.12	353.77	363.87	371.48	377.66	382.9	387.46	391.509	395.16
Nd	104.76	55.08	37.43	28.25	22.58	18.72	15.9	13.76	12.07
$^{87}\text{Sr}/^{86}\text{Sr}$	0.706082	0.705279	0.704799	0.704454	0.704183	0.703962	0.703774	0.703609	0.703463
$^{143}\text{Nd}/^{144}\text{Nd}$	0.512726	0.512747	0.512765	0.512787	0.512806	0.512825	0.512848	0.512872	0.512888
r = 0.2									
F	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
Sr	330.4	345.85	356.64	365.21	372.43	378.73	384.35	389.46	394.15
Nd	128.52	66.7	44.56	32.99	25.82	20.92	17.34	16.61	12.45
$^{87}\text{Sr}/^{86}\text{Sr}$	0.708766	0.707347	0.706427	0.705738	0.70518	0.70472	0.704298	0.703947	0.703626
$^{143}\text{Nd}/^{144}\text{Nd}$	0.512574	0.512602	0.512634	0.512665	0.512697	0.512734	0.512772	0.512822	0.512861
r = 0.25									
F	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
Sr	327.14	342.024	352.93	361.85	369.54	376.38	382.57	388.258	393.55
Nd	142.28	73.51	48.77	35.8	27.74	22.23	18.19	15.12	12.68
$^{87}\text{Sr}/^{86}\text{Sr}$	0.710038	0.708408	0.707301	0.706445	0.705742	0.705194	0.704614	0.704145	0.70372
$^{143}\text{Nd}/^{144}\text{Nd}$	0.512508	0.512539	0.512573	0.512607	0.512646	0.51269	0.512736	0.512787	0.512846
r = 0.35									
F	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
Sr	322.12	334.91	345.44	354.78	363.18	371.03	378.41	385.41	392.08
Nd	174.64	89.8	58.92	42.61	32.43	25.43	20.3	16.36	13.24
$^{87}\text{Sr}/^{86}\text{Sr}$	0.712333	0.710538	0.709162	0.708017	0.707024	0.706143	0.705349	0.704623	0.703954
$^{143}\text{Nd}/^{144}\text{Nd}$	0.512395	0.512427	0.512467	0.512501	0.512546	0.512596	0.512654	0.512728	0.51281

¹End member = C88-79 (tholeiitic basalt); $D_0\text{Sr} = 1.06$, $D_0\text{Nd} = 0.1$; contaminant = upper crust. CIPW norms for the starting composition are as follows: plg = 55.69; K-felds = 1.06; Ne = 2.81; Ol = 16.06; Cpx = 19.58. Trace and isotopic data for the starting composition are (C88-79): Sr = 398.5; Nd = 10.7 (ppm); $^{87}\text{Sr}/^{86}\text{Sr} = 0.703334$; $^{143}\text{Nd}/^{144}\text{Nd} = 0.512920$.

calculated D_0 values, and CIPW norms of the mafic end-member are given in Table 3. Various AFC trajectories based on different r (ratio of assimilation rate to crystallization rate) values are illustrated in the $^{87}\text{Sr}/^{86}\text{Sr}$ vs. $^{143}\text{Nd}/^{144}\text{Nd}$ diagram (Fig. 8). Calc-alkaline rocks plot toward the lower end of the most likely trajectories, but the samples do not fall on any of the trajectory lines. Figure 8 shows that the AFC process, starting from the tholeiitic end-member, appears to be partially responsible for development

of the calc-alkaline suite. This hypothesis is also supported by spidergrams that display similar trace-element patterns for calculated and measured chemical compositions (Table 4, Fig. 9).

Mixing between different source components

To shed further light on the origin of calc-alkaline lavas, two-component mixing (Langmuir et al., 1978) calculations were performed using selected source components. ERC96-145 and ERC98-1 were

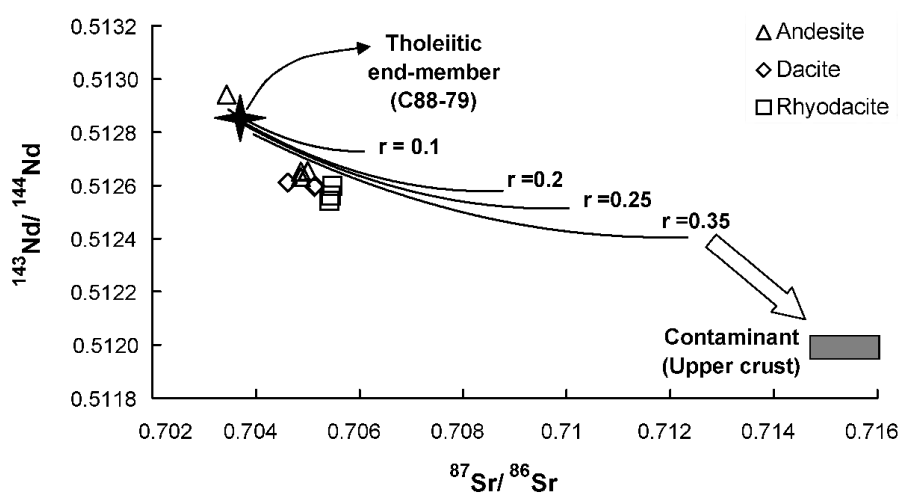


FIG. 8. Modeling of the AFC process starting from the tholeiitic end-member. End-member isotopic compositions are $^{87}\text{Sr}/^{86}\text{Sr} = 0.703334$; $^{143}\text{Nd}/^{144}\text{Nd} = 0.51292$. Contaminant parameters are given in the text; r = ratio of assimilation rate to fractionation rate.

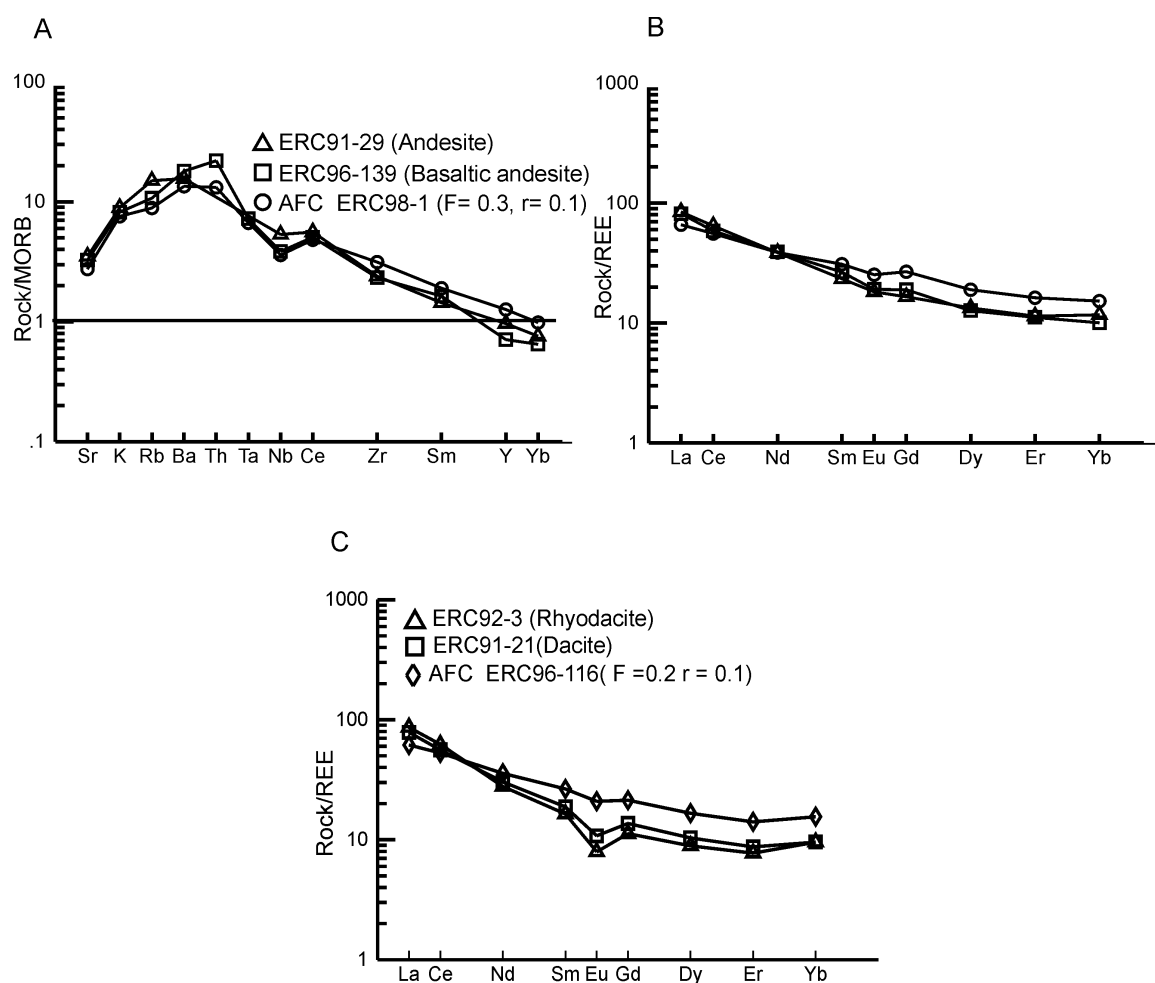


FIG. 9. MORB and chondrite-normalized spidergrams of trace elements calculated using AFC formulations (De Paolo, 1981) and the calc-alkaline products of the Erciyes volcanics. Calculation parameters are given in Table 3.

TABLE 4. Calculated Trace-Element Concentrations (ppm) from AFC Modeling and Some Measured Calc-alkaline Samples¹

	Calculated whole-rock partition coefficient (D_0)	Starting composition ²	Contaminant ³	Calculated trace- element composition ($F = 0.3$ $r = 0.1$)	Calculated trace element composition, ($F = 0.2$ $r = 0.1$)	ERC96-139	ERC91-29	ERC91-21
Rb (ppm)	0.061	5.97	112	17.75	10.43	21.44	30	
Sr	1.18	365.6	350	330.7	349.5	392.6	421.7	
Ba	0.256	160.5	550	269	203.7	359.9	312.3	
K, K ₂ O	0.107	0.57%	2.80%	1.13%	0.78%	1.23%	1.32%	
Zr	0.056	166.5	190	282.7	210.5	209.8	215.8	
Y	0.136	23.2	22	37.96	28.14	21.3	29	
Ti	0.186	1.7(%)	3000					
Nb	0.107	7.02	25	12.6	9.16	13.52	18.7	
La	0.208	13.3	30	21.78	16.6	26.9	27.9	25.9
Ce	0.204	29.5	64	48.24	36.78	50.7	55.9	48.5
Nd	0.229	17.8	26	24.4	21.72	24.7	24.3	19.1
Sm	0.25	4.7	4.5	6.28	5.64	5.4	4.75	3.81
Eu	0.403	1.57	0.88	1.95	1.79	1.47	1.4	0.83
Gd	0.273	5.64	3.8	7.4	6.69	5.23	4.57	3.77
Dy	0.211	4.83	3.5	6.5	5.82	4.35	4.59	3.55
Er	0.347	2.86	2.3	3.66	3.34	2.51	2.57	1.96
Yb	0.216	2.48	2.2	3.35	2.99	2.21	2.58	2.1

¹CIPW norms for the starting composition are as follows: Plg = 58.74; K-felds = 3.37; Ol = 14.92; Cpx = 12.03. r = ratio of assimilation rate to fractionation rate; F = remaining melt fraction.

²ERC98-1

³Upper crust after Taylor and Mc Lennan, 1985.

selected as the enriched and depleted end-members, because of their primitive compositions. The Zr/Nb – Y/Nb diagram (Fig. 10) shows two different mixing curves involving two distinct source components. Mixing curve 2 shows the standard interaction of MORB and OIB sources (data from Sun and McDonough, 1989), and is given for comparison with similar interacted source components. Of all the calc-alkaline products that plot along and around the mixing curve 1, ERC96-145 reflects an OIB-like whereas ERC98-1 reflects a depleted MORB-like source component. The only samples that can be attributed confidently to an asthenospheric mantle source are the Erciyes tholeiites. Although the mixing of partial melts derived from these asthenospheric sources had a significant influence on calc-alkaline development, the

involvement of lithospheric material during magma ascent cannot be ruled out. The effect of crustal contribution was also determined for the Karasu Valley Basaltic Volcanics (KVBV, southeast Turkey). Quartz and olivine tholeiitic lavas from the KVBV were generated from a common OIB-like source via varying degrees of crustal involvement (Alici et al., 2001). The development of lava products (KVBV) is demonstrated by AFC modeling and represented by various AFC trajectories (Alici et al., 2001). In addition, Figure 10 shows that the Colville volcanics (northwestern USA) also fall around the most likely mixing trajectory. Morris et al. (2000) suggested that calc-alkaline lavas of the Colville complex were generated from Proterozoic sources by crustal melting and that the LIL enrichment or the subduction signature were linked to Proterozoic crustal sources.

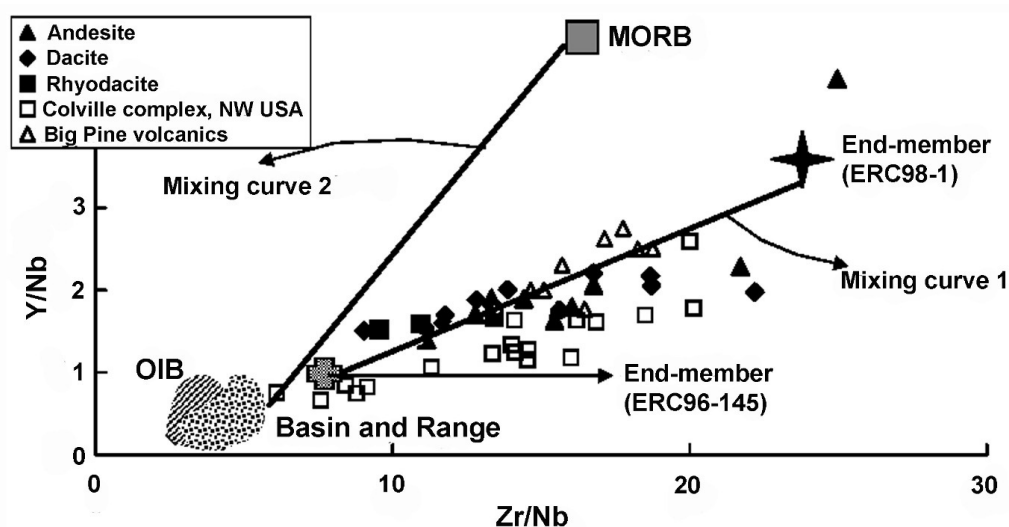


FIG. 10. Zr/Nb–Y/Nb diagram of calc-alkaline samples and magma mixing modeling between contrasted magma compositions. Data are from Kempton et al. (1987, 1991), Chaffey et al. (1989), Sun and McDonough (1989), and Woodhead (1996).

However, the Erciyes calc-alkaline lavas are younger than the Colville igneous complex. Incompatible-element enrichment of the mantle source may be inherited from fluids and melts that were released by pre-collisional subduction.

Thus, calc-alkaline lavas were generated by the mixing of partial melts of enriched and depleted asthenospheric sources. HFS element enrichment originated in the fluids and melts that were released from the source(s) by early subduction and AFC-related processes. The AFC model demonstrates that calc-alkaline rocks were generated from an initially tholeiitic basalt end-member via 20–30% assimilation of crustal material. REE patterns suggest that further siliceous products (rhyodacites) could have been generated by crustal melting (Fig 5D), but this is not widely supported by isotopic compositions. To summarize, calc-alkaline products were generated by the combined effects of the mixing of asthenospheric sources and a progressive AFC process that started from an initially tholeiitic source component in the absence of direct melting of the pre-subducting slab.

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